

This process is consistent with the MIMAS process currently being used to fabricate reactor-grade MOX fuel.

The nuclear design of the MOX fuel assembly has to account for differences in the nuclear characteristics between MOX and LEU fuel. The thermal absorption cross section for MOX is quite a bit larger than for LEU and results in lower neutron flux levels in MOX fuel assemblies compared to LEU fuel assemblies. This causes a large thermal neutron flux gradient at the MOX/LEU interfaces, which could result in high-power peaking factors in the outermost pins of the MOX fuel assemblies.

Also, the large thermal absorption and fission cross sections of MOX fuel results in a hardened neutron spectrum that reduces the effectiveness of the thermal neutron absorbers, namely, the soluble boron, the burnable poison (BP) rods, and the control rods. The differences between the two fuel types also produce different depletion behavior. The differences in the fuel characteristics are manageable through careful selection of assembly average plutonium enrichments and enrichment zoning within the MOX fuel assembly. Thus, neutronic differences between the MOX and LEU fuel can be minimized to lessen the perturbations associated with substituting one fuel type for the other.

2. SYSTEM DEFINITION

The design details needed for modeling the nuclear reactors at Catawba Nuclear Station Units 1 and 2 include materials and dimensions of reactor components and structures and operating conditions. These actual or assumed data are presented in Tables 1–6. The quantities are presented below both in “engineering” dimensional units and in the International System of Units (SI) or scientific units, as appropriate and necessary for use in computer code input.

Table 1. Reactor operating details and conditions for the Catawba Nuclear Station

Parameter	Value	SI (where appropriate)
Reactor design	Westinghouse PWR, 4-loop cooling system	
Thermal power	3411 MW(t)	
Power density	100.6 kW/L (core)	
Specific power level	38.7 kW/kgHM	
Average linear power/rod	5.44 kW/ft	17.848 kW m ⁻¹
Normal peak linear power/rod	13.58 kW/ft	44.554 kW m ⁻¹
Core diameter	132.7 in.	337.058 cm
Core “barrel”	ID: 148.0 in. OD: 152.5 in.	ID: 375.92 cm OD: 387.35 cm
Cross-sectional area of core	96.06 ft ²	8.924 m ²
Core height/diameter ratio	1.09	
Ratio of H ₂ O molecules to U atom	2.68–2.73	
Effective flow area for heat transfer	51.9 ft ²	4.822 m ²
Average heat flux	189400 Btu/h-ft ²	59.748 J s ⁻¹ cm ⁻²
Maximum heat flux (normal)	440370 Btu/h-ft ²	138.919 J s ⁻¹ cm ⁻²
Total thermal flow rate	1.412 × 10 ⁸ lb _m /h	1.779 × 10 ⁴ kg s ⁻¹
Effective flow rate for heat transfer	1.306 × 10 ⁸ lb _m /h	1.646 × 10 ⁴ kg s ⁻¹
Average coolant velocity along rods	15.166 ft/s	4.623 m s ⁻¹
Core coolant flow rate	1.336 × 10 ⁸ lb _m /h	1.6833 × 10 ⁴ kg s ⁻¹
Average mass velocity	2.516 × 10 ⁶ lb _m /h-ft ²	0.3413 kg s ⁻¹ cm ⁻²
Number of fuel assemblies in core	193	
Control rods	53	
Core inlet coolant temperature	Unit 1: 556.4°F Unit 2: 558.3°F	Unit 1: 291.33°C Unit 2: 292.39°C

Table 1. (continued)

Parameter	Value	SI (where appropriate)
Average coolant temperature in core	Unit 1: 586.7°F Unit 2: 589.3°F	Unit 1: 308.17°C Unit 2: 309.61°C
Average temperature rise in core	Unit 1: 60.7°F Unit 2: 61.9°F	Unit 1: 33.72°C Unit 2: 34.39°C
Minimum operating pressure	2250 psia	15.513 MPa
Nominal system pressure	2280 psia	15.720 MPa
Fraction of heat generated in the fuel	97.4%	
UO ₂ feed assemblies (40% MOX)	48	
MOX feed assemblies (40% MOX)	36	
Fuel cycle duration	495 d (~18 months)	
Assumed capacity factor	85%	

Table 2. Catawba fuel assembly data

Parameter	Value	SI (where appropriate)
Design	FCF Mark-BW, 17 × 17, canless	
Length	159.8 in.	405.8920 cm
Rod length	152.16 in.	386.4864 cm
Dimensions	8.425 in. × 8.425 in.	21.4 cm × 21.4 cm
Assembly pitch	8.466 in.	21.50364 cm
Total number of fuel rods in core	50,952	
Fuel rods per assembly	264	
Number of guide tube thimbles	24	
Number of instrument channels	1	
Rod pitch	0.496 in.	1.25984 cm
Cladding material	M5 or Zircaloy-4 (Zircaloy-4 ^a)	
Cladding OD	0.374 in.	0.94996 cm
Cladding ID	0.329 in.	0.83566 cm
Cladding thickness	0.0225 in. (0.024 ^a)	0.05715 cm (0.06096)
Cladding gas gap (radial)	0.00325 in.	0.00826 cm
Fuel pellet diameter	0.3225 in. (0.3195 ^a)	0.81915 cm (0.81153 cm)
Guide thimble OD	0.482 in.	1.22428 cm
Guide thimble ID	0.450 in.	1.14300 cm
Guide thimble material	Zircaloy-4	
Instrument thimble OD	Upper: 0.482 in. Lower: 0.429 in.	1.22428 cm 1.08966 cm
Instrument thimble ID	Upper: 0.450 in. Lower: 0.397 in.	1.14300 cm 1.00838 cm
Instrument thimble material	Zircaloy-4	
Grid spacer material per core	2 end grids: Inconel-718 (782 lb) 6 intermediate grids: Zircaloy-4 (2928 lb)	End: 354.71 kg Intermed: 1328.12 kg
Heavy metal (HM) loading/assembly	463.3 kg	
Active stack length (cold dimension)	144 in.	365.76 cm
Fuel pellet material	PuO ₂ + depleted UO ₂ (ceramic, sintered)	
Fuel pellet length	0.400 in. (chamfered) (Mk-BW)	1.016 cm
Volume reduction (pellet chamfer and dish)	1.0%	
Pellet theoretical density	95% (96% ^a)	
Weight of fuel if UO ₂	220,213 lb	99,887.0 kg
Cladding weight	56,841 lb (of Zircaloy-4)	25,782.7 kg

^aReported in the Catawba Final Safety Analysis Report (FSAR).

Table 3. Catawba IFBA pellet data

Parameter	Value	SI
IFBA absorber material	Zirconium diboride, ZrB_2 (enhanced to 30 wt % in ^{10}B to B)	
Absorber density	0.061 lb/in ³	1.688 g cm ⁻³
Coating thickness	1.575 mils	0.0040 cm

Table 4. Catawba burnable poison rod (BPR) data

Parameter	Value	SI
BPR absorber material	Boron carbide/alumina matrix, $Al_2O_3-B_4C$	
Boron content	Variable	
Rod OD	0.381 in.	0.96774 cm
Cladding material	Stainless steel, SS-304L	
Cladding thickness	0.025 in. ^a	0.06350 cm

^aThis value was estimated; no Catawba data were available.

Table 5. Catawba control rod data

Parameter	Value	SI
Control rod design	Hybrid B_4C	
Absorber material	B_4C	
Number of control rod clusters	53	
Number of absorber rods/cluster	24	
Absorber diameter	0.294 in.	0.74676 cm
Absorber density	0.064 lb/in. ³	1.7715 g cm ⁻³
Rod tip material	Ag-In-Cd (80 wt %, 15 wt %, 5 wt %)	
Tip diameter	0.301 in.	0.76454 cm
Tip length	40 in.	101.60 cm
Tip density	0.367 lb/in. ³	10.159 g cm ⁻³
Cladding material	Stainless steel 304L and 316, cold worked	
Cladding thickness	0.0385 in.	0.09779 cm
Full length dry weight (per assembly)	94 lb	42.638 kg
Absorber length	142 in.	360.68 cm

Table 6. Catawba reactor core baffle data

Parameter	Value	SI
Baffle material	Carbon steel	
Baffle thickness	1.125 in.	2.8575 cm